A Robust Digital Baseband Predistorter Constructed Using Memory Polynomials

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***Abstract*—Power amplifiers (PAs) are inherently nonlinear devices and are used in virtually all communications systems. Digital baseband predistortion is a highly cost-effective way to linearize PAs, but most existing architectures assume that the PA has a memoryless nonlinearity. For wider bandwidth applications such as wideband code-division multiple access (WCDMA) or wideband orthogonal frequency-division multiplexing (W-OFDM), PA memory effects can no longer be ignored, and memoryless predistortion has limited effectiveness. In this paper, instead of focusing on a particular PA model and building a corresponding predistorter, we focus directly on the predistorter structure. In particular, we propose a memory polynomial model for the predistorter and implement it using an indirect learning architecture. Linearization performance is demonstrated on a three-carrier WCDMA signal.**

***Index Terms*—Communication system nonlinearities, nonlinear systems, polynomials, power amplifiers (PAs), Volterra series.**

I. Introduction

POWER AMPLIFIERS (PAs) are indispensable components in a communication system and are inherently nonlinear. It is well known that there is an approximate inverse relationship between the PA efficiency and its linearity. Hence, nonlinear PAs are desirable from an efficiency point of view. The price paid for higher efficiency is that nonlinearity causes spectral regrowth (broadening), which leads to adjacent channel interference. It also causes in-band distortion, which degrades the bit-error rate (BER) performance. Newer transmission formats, such as code-division multiple access (CDMA) and orthogonal frequency-division multiplexing (OFDM), are especially vulnerable to PA nonlinearities, due to their high peak-to-average power ratio, corresponding to large fluctuations in their signal envelopes. In order to comply with spectral masks imposed by regulatory bodies and to reduce BER, PA linearization is necessary.

Paper approved by Z. Kostic, the Editor for Wireless Communication of the IEEE Communications Society. Manuscript received July 17, 2002; revised May 22, 2003. This work was supported in part by the National Science Foundation under Grant 9703312 and Grant 0219262, and in part by the State of Georgia's Yamacraw Initiative. This paper was presented in part at the IEEE Global Telecommunications Conference, Taipei, Taiwan, November, 2002.

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Digital Object Identifier 10.1109/TCOMM.2003.822188

Of all linearization techniques, digital baseband predistortion is among the most cost effective. A predistorter is a functional block that precedes the PA. It generally creates an expanding nonlinearity since the PA has a compressing characteristic. Ideally, we would like the PA output to be a scalar multiple of the input to the predistorter-PA chain. For a memoryless PA (i.e., the current output depends only on the current input), memoryless predistortion is sufficient. There has been intensive research on memoryless predistortion during the past decade [1].

For wider bandwidth applications such as wideband CDMA (WCDMA) or wideband OFDM (W-OFDM), PA memory effects can no longer be ignored. Moreover, higher power amplifiers such as those used in wireless base stations exhibit memory effects. The cause of memory effects can be electrical or electrothermal as suggested in [2]. Memoryless predistortion for a PA with memory often results in poor linearization performance.

Most researchers approach the problem of PA linearization by first finding a good model for the PA. Volterra series is a general nonlinear model with memory [3] and has been used to model PAs with mild nonlinearities [4]. A serious drawback of the Volterra model is the large number of coefficients that must be extracted. Predistortion of the Volterra model is usually implemented using the pth-order inverse technique [3], which is complicated. One thing to keep in mind is that the exact inverse of a Volterra system is difficult to construct and the pth-order inverse is only an approximation. Recently, two special cases of the Volterra model have been proposed to capture the memory nonlinear effects in the PA associated with wideband signals. One is the Wiener model; i.e., a linear time-invariant (LTI) system followed by a memoryless nonlinearity, proposed by Clark *et al.* [5]. The advantage of Wiener modeling for the PA is that the corresponding predistorter is a Hammerstein system; i.e., a memoryless nonlinearity followed by an LTI system, and it is possible for the predistorter to be an exact inverse of the PA. Another special case of the Volterra model is the memory polynomial model proposed by Kim *et al.* [6]. Similar to the Volterra model, an exact inverse of the memory polynomial is difficult to obtain, but another memory polynomial can be constructed as an approximate inverse. A variety of other PA models exist, for example, the parallel Wiener model investigated in [7].

It is difficult to judge which PA model is the best, since it could depend on the type of the PA, the data format being transmitted, etc. Moreover, the most accurate PA model may not be the most amenable to predistortion. In applications that we are interested in, predistortion linearization is the ultimate objective, whereas accurate PA modeling is only a secondary concern.

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For a nonlinear PA with memory, its inverse must also be a nonlinear system with memory. Our goal here is to find a good model to approximate the inverse of the PA nonlinearity, in the sense that parameter extraction and system implementation are straightforward and the predistorter model is robust. In [6], it was shown that a memory polynomial is a good model for the PA, so it is then natural for us to consider it for the predistorter as well.

In this paper, we impose a memory polynomial model on the predistorter and implement it using the indirect learning architecture proposed in [8]. Simulation results show that this is an improvement over our previous work [9] where a Hammerstein predistorter is used. The improvements are mostly in terms of robustness and simplicity of predistorter parameter estimation.

II. Memory Polynomial Model

Let us consider, as an example, a baseband linear-cubic model with input $x(n)$ and output $z(n)$

$$z(n) = \sum_{q=0}^{Q} h_1(q)x(n-q) + \sum_{q_1,q_2,q_3=0}^{Q} h_3(q_1,q_2,q_3) \times x(n-q_1)x(n-q_2)x^*(n-q_3) \quad (1)$$

where the linear and cubic kernels $\{h_1(q)\}_{q=0}^{Q}$ and $\{h_3\{q_1,q_2,q_3)\}_{q_1,q_2,q_3=0}^{Q}$ characterize the nonlinear system. If $Q=0$, then (1) reduces to

$$z(n) = h_1(0)x(n) + h_3(0,0,0)x(n)\,|x(n)|^2 \quad (2)$$

which is a memoryless nonlinear system. If $h_3(q_1,q_2,q_3)=0$, except along the diagonal $q_1=q_2=q_3$, then (1) becomes

$$z(n)=\sum_{q=0}^{Q}\left[h_1(q)x(n-q)+h_3(q,q,q)x(n-q)\,|x(n-q)|^2\right] \quad (3)$$

and we refer to (3) as a memory polynomial. We can view the memory polynomial as a compromise between the memoryless nonlinearity and the full Volterra nonlinearity. It has the advantage of capturing memory effects while keeping the number of coefficients on the order of $Q+1$. This is contrasted with the full Volterra system, where the number of unknowns is on the order of $(Q+1)^K$, where K is the nonlinearity order.

III. Indirect Learning Architecture

Fig. 1 shows the indirect learning structure that is used for predistorter identification [8]. The feedback path labeled "Predistorter Training" (block A) has $y(n)/G$ as its input, where G is the gain of the linearized PA, and $\hat{z}(n)$ as its output. The actual predistorter is an exact copy of the feedback path (copy of A); it has $x(n)$ as its input and $z(n)$ as its output. Ideally, we would like $y(n)=Gx(n)$, which renders $z(n)=\hat{z}(n)$ and the error term $e(n)=0$. Given $y(n)$ and $z(n)$, our task is to find the parameters of block A, which yields the predistorter. The algorithm converges when the error energy $||e(n)||^2$ is minimized.

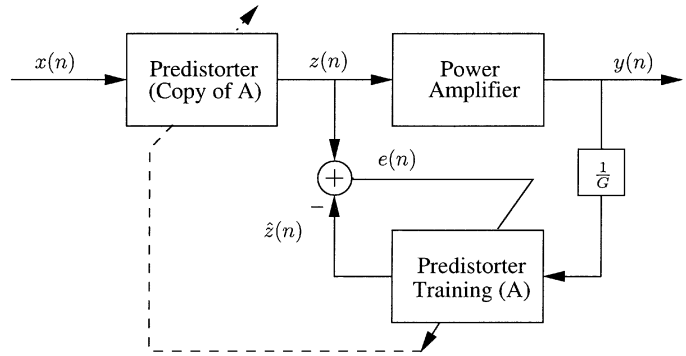

Fig. 1. Indirect learning architecture for the predistorter.

We need to assume that the PA nonlinearity is invertible so that $e(n)=0$ corresponds to $y(n)=Gx(n)$.

Here, we consider that the PA characteristics do not change rapidly with time; changes in PA characteristics are often due to temperature drift, aging, etc., which have long time constants. After gathering a block of $y(n)$ and $z(n)$ data samples, the training branch (block A) can process the data offline, which lowers the processing requirements of the predistortion system. Once the predistorter identification algorithm has converged, the new set of parameters are plugged into the high-speed predistorter, which can be readily implemented using application-specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs). When the predistorter coefficients have been found and it is believed that the PA characteristics are hardly changing, the setup in Fig. 1 can be run in open loop, i.e., we temporarily shut down the training branch, until changes in the PA characteristics require a predistorter coefficient update.

IV. Predistorter Identification

Generalizing (3) and changing $h_k(q,\ldots,q)$ to a_{kq}, the memory polynomial predistorter (similar to [6]) can be described by

$$z(n) = \sum_{k=1}^{K}\sum_{q=0}^{Q} a_{kq}\, x(n-q)\,|x(n-q)|^{k-1} \quad (4)$$

where we have slightly generalized the model by including even- as well as odd-order terms. Therefore, the predistorter has memory Q and highest nonlinearity order K. In most predistorter designs, only odd-order nonlinearities are included, i.e., $k=2l+1$, and the term $x(n-q)|x(n-q)|^{k-1}=[x(n-q)]^{l+1}[x^*(n-q)]^l$ is of order $2l+1$. Thus, the exponent of $x(n-q)$ is $l+1$, and the exponent of $x^*(n-q)$ is l—such designation follows the analysis of [10, p. 69]. As we will show in *Example 2* (Section V), by including even-order nonlinear terms in the predistorter [i.e., allow even k in (4)], spectral regrowth can be further reduced (by about 3–5 dB in *Example 2*). Detailed investigation of the benefits of even-order terms in the baseband model (4) is reported in [11]. Note that the even-order term $x(n-q)|x(n-q)|^{k-1}$ is different from $|x(n-q)|^k$. The former contains phase information, whereas the latter is "phase blind."

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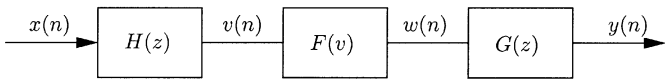

Fig. 2. W–H model diagram.

Since $z(n)$ is linear in the parameters a_{kq}, the latter can be estimated by a simple least-squares method. By defining a new sequence

$$u_{kq}(n) = \frac{y(n-q)}{G}\left|\frac{y(n-q)}{G}\right|^{k-1} \tag{5}$$

at convergence, we should have

$$\mathbf{z} = \mathbf{U}\mathbf{a} \tag{6}$$

where $\mathbf{z} = [z(0), \ldots, z(N-1)]^T$, $\mathbf{U} = [\mathbf{u}_{10}, \ldots, \mathbf{u}_{K0}, \ldots, \mathbf{u}_{1Q}, \ldots, \mathbf{u}_{KQ}]$, $\mathbf{u}_{kq} = [u_{kq}(0), \ldots, u_{kq}(N-1)]^T$, and $\mathbf{a} = [a_{10}, \ldots, a_{K0}, \ldots, a_{1Q}, \ldots, a_{KQ}]^T$. The least-squares solution for (6) is

$$\hat{\mathbf{a}} = (\mathbf{U}^H\mathbf{U})^{-1}\mathbf{U}^H\mathbf{z} \tag{7}$$

where $(\cdot)^H$ denotes complex conjugate transpose.

V. Simulations

In this section, we illustrate, through computer simulations, performance of the memory polynomial predistorter identified using the indirect learning architecture. We will show that the same memory polynomial structure can be used to linearize several different nonlinear models with memory, hence, demonstrating the robustness of memory polynomial predistortion.

Example 1: Here, the nonlinearity to be compensated for is assumed to obey a Wiener–Hammerstein (W–H) model (see Fig. 2), i.e., an LTI system followed by a memoryless nonlinearity, which in turn is followed by another LTI system. Such a configuration is commonly used for satellite communication channels where the PA at the satellite transponder is driven near saturation to exploit the maximum power efficiency [12]. The LTI blocks before and after the memoryless nonlinearity, which are denoted by $H(z)$ and $G(z)$, respectively, are assumed to be

$$H(z) = \frac{1+0.5z^{-2}}{1-0.2z^{-1}} \quad G(z) = \frac{1-0.1z^{-2}}{1-0.4z^{-1}}. \tag{8}$$

For the memoryless nonlinear portion of the W–H model

$$w(n) = \sum_{\substack{k=1\\ k \text{ odd}}}^{K} b_k v(n)\,|v(n)|^{k-1} \tag{9}$$

where $v(n)$ and $w(n)$ are, respectively, input and output of the memoryless nonlinear block. For the coefficients, we had

$$\begin{aligned} b_1 &= 1.0108 + 0.0858j \\ b_3 &= 0.0879 - 0.1583j \\ b_5 &= -1.0992 - 0.8891j \end{aligned} \tag{10}$$

which were extracted from an actual Class AB PA.

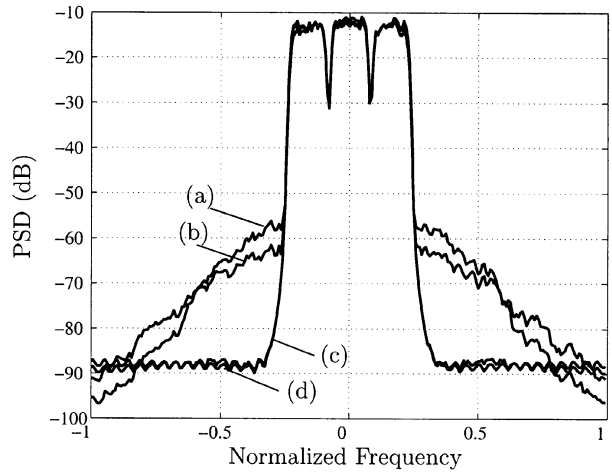

Fig. 3. Effectiveness of predistortion in suppressing spectral regrowth when the PA is modeled by a W–H system. (a) Output without predistortion. (b) Output with memoryless predistortion. (c) Output with memory polynomial predistortion ($Q = 2$, $K = 5$). (d) Original input. (c) and (d) almost coincide.

The baseband input is a three-carrier WCDMA signal. Memory polynomial predistorter identification is carried out based on 8000 data samples. Next, we compare the power spectral density (PSD) of the input and output signals to evaluate the effectiveness of the predistorter in reducing spectral regrowth. Here, we used the predistorter (4) with two delay taps ($Q = 2$) and fifth odd-order nonlinearity ($K = 5$). Performance of the predistorter is demonstrated in Fig. 3. Spectral regrowth is almost fully suppressed with only two delay taps, even though the LTI portions of the W–H system have much longer impulse responses.

Example 2: Here, we assume that the PA obeys an odd-order-only memory polynomial model

$$y(n) = \sum_{\substack{k=1\\ k \text{ odd}}}^{K}\sum_{q=0}^{Q} c_{kq}\, z(n-q)\,|z(n-q)|^{k-1}. \tag{11}$$

The coefficients

$$\begin{aligned} c_{10} &= 1.0513 + 0.0904j & c_{30} &= -0.0542 - 0.2900j \\ c_{50} &= -0.9657 - 0.7028j & c_{11} &= -0.0680 - 0.0023j \\ c_{31} &= 0.2234 + 0.2317j & c_{51} &= -0.2451 - 0.3735j \\ c_{12} &= 0.0289 - 0.0054j & c_{32} &= -0.0621 - 0.0932j \\ c_{52} &= 0.1229 + 0.1508j \end{aligned} \tag{12}$$

were extracted from the same PA as in *Example 1*. Fig. 4 shows the performance of various predistorters. The memory polynomial predistorter with $Q = 2$ and $K = 5$ was able to suppress most of the spectral regrowth. However, when both even- and odd-order nonlinearities are included in the predistorter, an additional 3–5 dB suppression can be achieved. Spectral regrowth can be further suppressed by increasing the memory of the predistorter to $Q = 10$.

Example 3: We start with a Wiener model whose $H(z) = 1+0.5z^{-2}$, $b_1 = 1.0108+0.0858j$, and $b_3 = 0.0879-0.1583j$. Recall that the Wiener model can be regarded as a special case of the Volterra system. We added zero-mean complex Gaussian noise with variance 2×10^{-4} to the corresponding Volterra kernels $\{h_1(q)\}$, $\{h_3(q_1, q_2, q_3)\}$, and used the resulting Volterra

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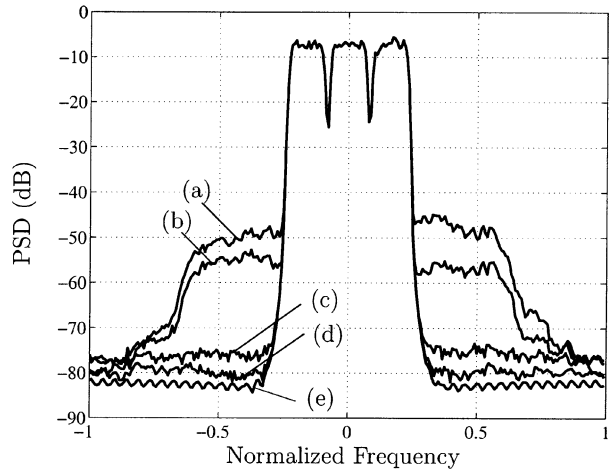

Fig. 4. Effectiveness of predistortion in suppressing spectral regrowth, when the PA itself is modeled by a memory polynomial. (a) Output without predistortion. (b) Output with memoryless predistortion. (c) Output with memory polynomial predistortion ($Q = 2$, $K = 5$, odd order). (d) Output with memory polynomial predistortion ($Q = 2$, $K = 5$, even and odd orders). (e) Original input.

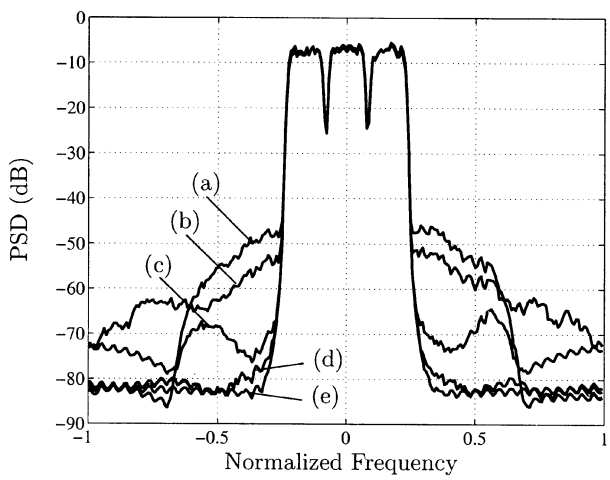

Fig. 5. Effectiveness of predistortion in suppressing spectral regrowth, when the PA is modeled by a perturbed Wiener (hence Volterra) system. (a) Output without predistortion. (b) Output with memoryless predistortion. (c) Output with memory polynomial predistortion ($Q = 2$, $K = 5$). (d) Output with memory polynomial predistortion ($Q = 10$, $K = 5$). (e) Original input.

system (perturbed Wiener) as the model for the PA. The results are shown in Fig. 5. We still observe significant reduction in spectral regrowth with the memory polynomial predistorter ($Q = 2$, $K = 5$). With the maximum delay increased to $Q = 10$, the predistorter almost fully suppressed the spectral regrowth.

Example 4: The PA here is assumed to follow a three-branch parallel Wiener model (sum of Wiener subsystems; see Fig. 6). The LTI blocks in the model are defined by

$$\begin{aligned} H_1(z) &= 1 \\ H_2(z) &= \frac{1+0.3z^{-1}}{1-0.1z^{-1}} \\ H_3(z) &= \frac{1-0.2z^{-1}}{1-0.4z^{-1}}. \end{aligned} \tag{13}$$

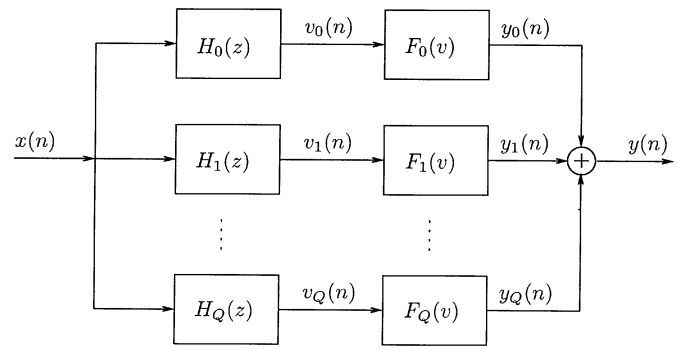

Fig. 6. Parallel Wiener model diagram. $H_i(\cdot)$ is an LTI block, and $F_i(\cdot)$ is a memoryless nonlinear block.

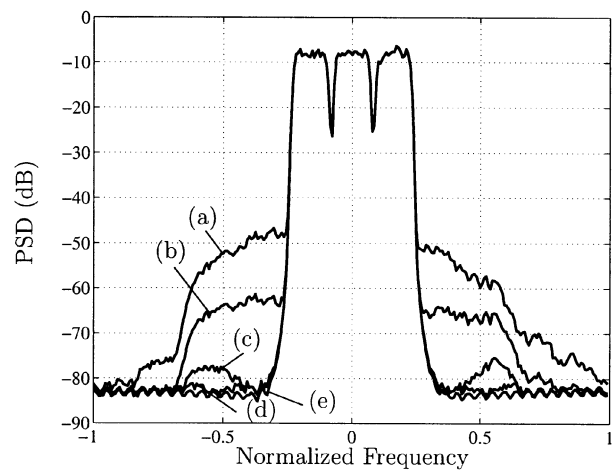

Fig. 7. Effectiveness of predistortion in suppressing spectral regrowth, when the PA is modeled by a parallel Wiener system. (a) Output without predistortion. (b) Output with memoryless predistortion. (c) Output with memory polynomial predistortion ($Q = 2$, $K = 5$). (d) Output with memory polynomial predistortion ($Q = 5$, $K = 5$). (e) Original input.

The memoryless nonlinearity in the ith branch has input/output relationship

$$y_i(n) = \sum_{\substack{k=1 \\ k \text{ odd}}}^{K} d_{ki}\, v_i(n)\, |v_i(n)|^{k-1} \tag{14}$$

where $v_i(n)$ and $y_i(n)$ are the input and output of the nonlinearity $F_i(v)$, respectively. The d_{ki} coefficients used were

$$\begin{aligned} d_{11} &= 1.0108 + 0.0858j & d_{31} &= 0.0879 - 0.1583j \\ d_{51} &= -1.0992 - 0.8891j & d_{12} &= 0.1179 + 0.0004j \\ d_{32} &= -0.1818 + 0.0391j & d_{52} &= 0.1684 + 0.0034j \\ d_{13} &= 0.0473 - 0.0058j & d_{33} &= 0.0395 + 0.0283j \\ d_{53} &= -0.1015 - 0.0196j. \end{aligned} \tag{15}$$

Since $H_1(z) = 1$, the first branch is actually a memoryless nonlinearity here. This reflects some belief that the dominating type of nonlinearity in a PA is memoryless. The second and third branches both exhibit memory nonlinearity, with 10 dB and 13 dB less power than the first branch, respectively.

Fig. 7 shows the performance of our memory polynomial predistorter in linearizing such a PA. With the memory polynomial predistorter ($Q = 2$, $K = 5$), there is a significant decrease in

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spectral regrowth, and the result is further improved when Q is increased to five.

In all of the above cases, memoryless predistortion is not very effective in suppressing spectral regrowth, which underscores the notion that PA memory effects must be taken into account when designing the predistorter.

The objective of PA linearization is two-fold: suppression of spectral regrowth to reduce adjacent channel interference and minimization of in-band distortion to improve BER. Although only PSD plots are shown here, this does not mean that in-band distortion is left unchecked. Recall that in the indirect learning architecture, our convergence criterion requires the mean squared error between $y(n)$ and $Gx(n)$ to be minimized. Therefore, at convergence, the PA is linearized, which automatically ensures the suppression of both in-band and out-of-band distortions. The PSD plots are shown for verification purposes. Because the PSD is phase blind, if one were to define a linearization criterion solely in terms of the PSD, the resulting predistorter may not be a true linearizer.

VI. Discussion

The Volterra series is the most general polynomial type of nonlinearity with memory. In this paper, we have encountered the memory polynomial, the Wiener, the Hammerstein, and the parallel Wiener models as special cases of the Volterra model. Next, we would like to point out some interesting links among these models.

For the memory polynomial model (4), let us collect the coefficients in a matrix

$$\mathbf{A} = \begin{bmatrix} a_{10} & a_{11} & \cdots & a_{1Q} \\ a_{30} & a_{31} & \cdots & a_{3Q} \\ \vdots & \vdots & \ddots & \vdots \\ a_{K0} & a_{K1} & \cdots & a_{KQ} \end{bmatrix}. \tag{16}$$

A Hammerstein model on the other hand, can be described by

$$v(n) = \sum_{k=1}^{K} c_k \, x(n) \, |x(n)|^{k-1} \tag{17}$$

$$y(n) = \sum_{q=0}^{Q} h(q) \, v(n-q). \tag{18}$$

Let us collect the coefficients in (17) and (18) in vectors $\mathbf{c} = [c_1, \ldots, c_K]^T$, $\mathbf{h} = [h(0), h(1), \ldots, h(Q)]^T$. Substitution of (17) into (18) yields

$$y(n) = \sum_{k=1}^{K} \sum_{q=0}^{Q} c_k \, h(q) \, x(n-q) \, |x(n-q)|^{k-1}. \tag{19}$$

Comparing (19) with (4), we can see that the Hammerstein system (17)–(18) is a special case of the memory polynomial model (4) with $a_{kq} = c_k h(q)$. In other words, $\mathbf{A} = \mathbf{c}\,\mathbf{h}^T$, and hence, rank $(\mathbf{A}) = 1$. This implies that as a predistorter, the memory polynomial is expected to work well with a Wiener PA. On the other hand, the memory polynomial predistorter is expected to be more robust than the Hammerstein predistorter. Interestingly, although the memory polynomial model is more general than the Hammerstein model, its parameter estimation is actually more straightforward [c.f., via the least-squares solution (7)].

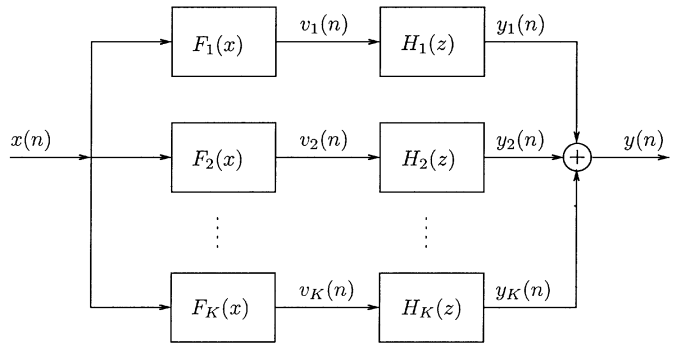

Fig. 8. Parallel Hammerstein model diagram. $F_i(\cdot)$ is a memoryless nonlinear block, and $H_i(\cdot)$ is an LTI block.

Let

$$v_k(n) = F_k\left(x(n)\right) = x(n) \, |x(n)|^{k-1} \tag{20}$$

$$y_k(n) = v_k(n) * h_k(n) \tag{21}$$

where $h_k(n) = a_{kn}$, and $*$ denotes convolution. We can rewrite the memory polynomial model as

$$y(n) = \sum_{k=1}^{K} y_k(n) = \sum_{k=1}^{K} v_k(n) * a_{kn} \tag{22}$$

$$= \sum_{k=1}^{K} \sum_{q=0}^{Q} a_{kq} \, x(n-q) \, |x(n-q)|^{k-1}. \tag{23}$$

Therefore, a memory polynomial model is also a parallel Hammerstein model (see Fig. 8) where the memoryless nonlinear block is a polynomial.

In [13], the authors tried to linearize a Wiener system with a Hammerstein predistorter using the indirect learning architecture. They adopted the least-squares approach[1] to solve for the predistorter coefficients, although the parameters of the memoryless nonlinear and the LTI blocks of the Hammerstein model are not explicitly recovered.

Alternatively, we can also rewrite the memory polynomial model as

$$v_q(n) = x(n) * h_q(n) \quad h_q(n) = \delta(n-q) \tag{24}$$

$$F_q(n) = \sum_{k=1}^{K} a_{kq} \, v|v|^{k-1} \tag{25}$$

$$y(n) = \sum_{q=0}^{Q} F_q\left[v_q(n)\right] \tag{26}$$

where $*$ in (24) denotes convolution. Comparing with the parallel Wiener model (see Fig. 6), we observe that a memory polynomial is also a special parallel Wiener model with

$$H_q(z) = z^{-q} \quad \text{or} \quad h_q(n) = \delta(n-q). \tag{27}$$

In summary, when considering the polynomial type of nonlinearities, both the parallel Wiener and parallel Hammerstein models are special cases of the Volterra series. In

[1]There is a typo in [13, eq. (13)]: $|x[n]|^2 x^*[n-1]$ should be $|x[n-1]|^2 x^*[n-1]$; $|x[n]|^2 x^*[n-2]$ should be $|x[n-2]|^2 x^*[n-2]$. Moreover, we believe that the baseband expression (9) should be in terms of $x[n-i]|x[n-i]|^{j-1}$ instead of $x^j[n-i]$.

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fact, it can be shown that the memory polynomial model is equivalent to the parallel Hammerstein model. We have also shown that a memory polynomial model is a special case of the parallel Wiener model. Obviously, the parallel Hammerstein model includes the Hammerstein model as a special case, and the parallel Wiener model includes the Wiener model as a special case. Hammerstein and Wiener models are the most "specialized" with the least number of coefficients, but are by no means the easiest to identify. The memory polynomial model, however, offers a good compromise between generality and ease of parameter estimation and implementation.

VII. Conclusions

In this paper, instead of trying to find a "good" memory nonlinear PA model and then devising a suitable predistorter for that particular model, we have focused directly on predistorter design. In particular, we adopted a memory polynomial structure for the predistorter and implemented it using an indirect learning architecture. The predistorter parameters are easy to extract, involving only linear least squares. We demonstrated that such a predistorter is not tied to a particular PA model and is therefore robust. The effectiveness of predistortion is demonstrated on a W–H system, a memory polynomial nonlinearity, a perturbed Wiener (full Volterra) system, and a parallel Wiener model.

Acknowledgment

The authors also wish to thank R. Raich for many insightful discussions on the topic of predistortion linearization of PAs with memory effects.

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